

Mathematical Constructs of Alternative CMB Models:
Unnikrishnan and Peratt in Harmony with T0 Theory
A Detailed Analysis of the Field Equations and Their Synthesis with the ξ -Field

Contents

.1	Introduction: From Surface to Mathematical Analysis	1
.2	Mathematical Constructs of Cosmic Relativity (Unnikrishnan)	1
	Fundamental Field Equations	1
	CMB Derivation: Standing Waves	2
.3	Mathematical Constructs of Plasma Cosmology (Peratt)	2
	Fundamental Field Equations	2
	CMB Derivation: Spectrum and Power Spectrum	2
.4	Synthesis: Harmony with T0 Theory	3
	Unnikrishnan in T0	3
	Peratt in T0	3
	Unified T0 Equation	3
.5	Conclusion	3

Abstract

Based on the video “The CMB Power Spectrum – Cosmology’s Untouchable Curve?”, we analyze in detail the mathematical foundations of the alternative models proposed by C. S. Unnikrishnan (cosmic relativity) and Anthony L. Peratt (plasma cosmology). Unnikrishnan’s field equations extend special relativity by incorporating universal gravitational effects within a static space, while Peratt’s Maxwell-based plasma model derives the CMB from synchrotron radiation. We demonstrate how both constructs are compatible with T0 theory: the ξ -field ($\xi = \frac{4}{3} \times 10^{-4}$) serves as a universal parameter that unifies resonance modes (Unnikrishnan) and filament dynamics (Peratt). The resulting synthesis yields a coherent, expansion-free cosmology in which the CMB power spectrum is explained as an emergent ξ -harmony.

.1 Introduction: From Surface to Mathematical Analysis

The video [5] highlights the circular nature of the Λ CDM model and contrasts it with radical alternatives: Unnikrishnan’s static resonance and Peratt’s plasma-based radiation. A superficial view is insufficient; we delve deeply into the field equations and derivations, based on primary sources [1, 2]. The goal is a synthesis with T0 theory, where the ξ -field connects the time–mass duality ($T \cdot m = 1$) and fractal geometry. This resolves open issues such as the high Q-factor and spectral precision.

.2 Mathematical Constructs of Cosmic Relativity (Unnikrishnan)

Unnikrishnan's theory [1] reformulates relativity as "cosmic relativity": relativistic effects are gravitational gradients in a homogeneous, static universe. No expansion; CMB peaks arise as standing waves in a cosmic field.

Fundamental Field Equations

The core idea: Lorentz transformations $L(v, t)$ become gravitational effects:

$$L(v, t) = \exp\left(-\frac{\nabla\Phi}{c^2}\right), \quad (1)$$

where Φ is the cosmic gravitational potential ($\Phi = -GM/r$ for a homogeneous universe, M = total mass). Time dilation and length contraction emerge as:

$$\frac{\Delta t}{t} = 1 + \frac{\Phi}{c^2}, \quad \frac{\Delta l}{l} = 1 - \frac{\Phi}{c^2}. \quad (2)$$

The field equation extends Einstein's equations to a "cosmic metric":

$$R_{\mu\nu} = 8\pi G\left(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T\right) + \Lambda g_{\mu\nu} + \xi \nabla_\mu \nabla_\nu \Phi, \quad (3)$$

with ξ as the coupling constant (here analogous to T_0). The Weyl part $W_{\mu\nu\rho\sigma}$ represents anisotropic cosmic gradients.

CMB Derivation: Standing Waves

CMB as resonance modes in a static field. The wave equation in the cosmic frame:

$$\square\psi + \frac{\nabla\Phi}{c^2}\partial_t\psi = 0, \quad (4)$$

leads to standing waves $\psi = \sum_k A_k \sin(k \cdot x - \omega t + \phi_k)$, with peaks at $k_n = n\pi/L_{\text{cosmic}}$ (L = cosmic size). Q-factor $Q = \omega/\Delta\omega \approx 10^6$ due to gravitational damping. Polarization arises from W -induced phase shifts.

The video (11:46) describes this as "living resonance" – mathematically: harmonic oscillators in Φ -gradients.

.3 Mathematical Constructs of Plasma Cosmology (Peratt)

Peratt's model [2] derives the CMB from plasma dynamics: synchrotron radiation in Birkeland filaments produces a blackbody spectrum through collective emission/absorption.

Fundamental Field Equations

Based on Maxwell's equations in plasmas:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \quad \nabla \cdot \mathbf{B} = 0, \quad (5)$$

with Lorentz force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. For filaments: Z-pinch equation

$$\frac{dp}{dt} = \mathbf{J} \times \mathbf{B}, \quad (6)$$

where $\mathbf{J}$ is current density (10^{18} A in galactic filaments). Synchrotron power:

$$P_{\text{synch}} = \frac{2}{3} r_e^2 \gamma^2 \beta^2 c B_{\perp}^2 \sin^2 \theta, \quad (7)$$

with r_e classical electron radius, γ Lorentz factor.

CMB Derivation: Spectrum and Power Spectrum

Collective radiation: integrated spectrum over N filaments:

$$I(\nu) = \int N(\mathbf{r}) P_{\text{synch}}(\nu, B(\mathbf{r})) e^{-\tau(\nu)} d\mathbf{r}, \quad (8)$$

where $\tau(\nu)$ is optical depth (self-absorption). For CMB fit: $T \approx 2.7$ K at $\nu \approx 160$ GHz; peaks as interference:

$$C_{\ell} = \frac{1}{2\ell + 1} \sum_m |a_{\ell m}|^2, \quad a_{\ell m} \propto \int Y_{\ell m}^*(\theta, \phi) e^{i\mathbf{k} \cdot \mathbf{r}} d\Omega, \quad (9)$$

with $\mathbf{k}$ wave vector in filament magnetic fields. BAO: fractal scales $r_n = r_0 \phi^n$ (ϕ golden ratio).

The video (13:46) emphasizes "pure electrodynamics" – Peratt's simulations match the SED to within 1%.

.4 Synthesis: Harmony with T0 Theory

T0 unifies both approaches via the ξ -field: a static universe with fractal geometry, where redshift arises as a geometric path effect in the ξ -field (see Document 026 for the quantitative derivation).

Unnikrishnan in T0

ξ as cosmic coupling parameter: replaces $\nabla\Phi/c^2$ with $\xi\nabla \ln \rho_{\xi}$, where ρ_{ξ} is ξ -density. Extended equation:

$$R_{\mu\nu} = 8\pi G T_{\mu\nu} + \xi \nabla_{\mu} \nabla_{\nu} \ln \rho_{\xi}. \quad (10)$$

Resonance modes: $\square\psi + \xi \mathcal{F}[\psi] = 0$ (T0 field equation), peaks at $\omega_n = nc/L \cdot (1 - 100\xi)$. Q-factor: $Q \approx 1/(1 - K_{\text{frak}}) = 1/(100\xi) = 75$.

Peratt in T0

Filaments as ξ -induced currents: $\mathbf{J} = \sigma \mathbf{E} + \xi \nabla \times \mathbf{B}$. Synchrotron:

$$P_{\text{synch}} = \frac{2}{3} r_e^2 \gamma^2 \beta^2 c (B_{\perp} + \xi \partial_t B)^2. \quad (11)$$

Power spectrum: fractal hierarchy $C_{\ell} \propto \sum_n \xi^n \sin(\ell \theta_n)$, with $\theta_n = \pi(1 - 100\xi)^n$. BAO: $r_{\text{BAO}} \approx 150$ Mpc as ξ -scaled filament length.

Unified T0 Equation

Combined field equation:

$$\square A_{\mu} + \xi (\nabla^{\nu} F_{\nu\mu} + \mathcal{F}[A_{\mu}]) = J_{\mu}, \quad (12)$$

where A_{μ} is the vector potential (Peratt), $\mathcal{F}$ the fractal operator (Unnikrishnan/T0). This generates the CMB as ξ -resonance in a static plasma field.

.5 Conclusion

The mathematical constructs of Unnikrishnan (gravitational Lorentz transformations) and Peratt (Maxwell–synchrotron in filaments) are coherent yet isolated. T0 brings them into harmony: ξ serves as the bridge between resonance and plasma dynamics. The CMB power spectrum emerges as ξ -harmony – precise and without ad-hoc patches. Future simulations (e.g. FEniCS for ξ -fields) will provide further tests.

Bibliography

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